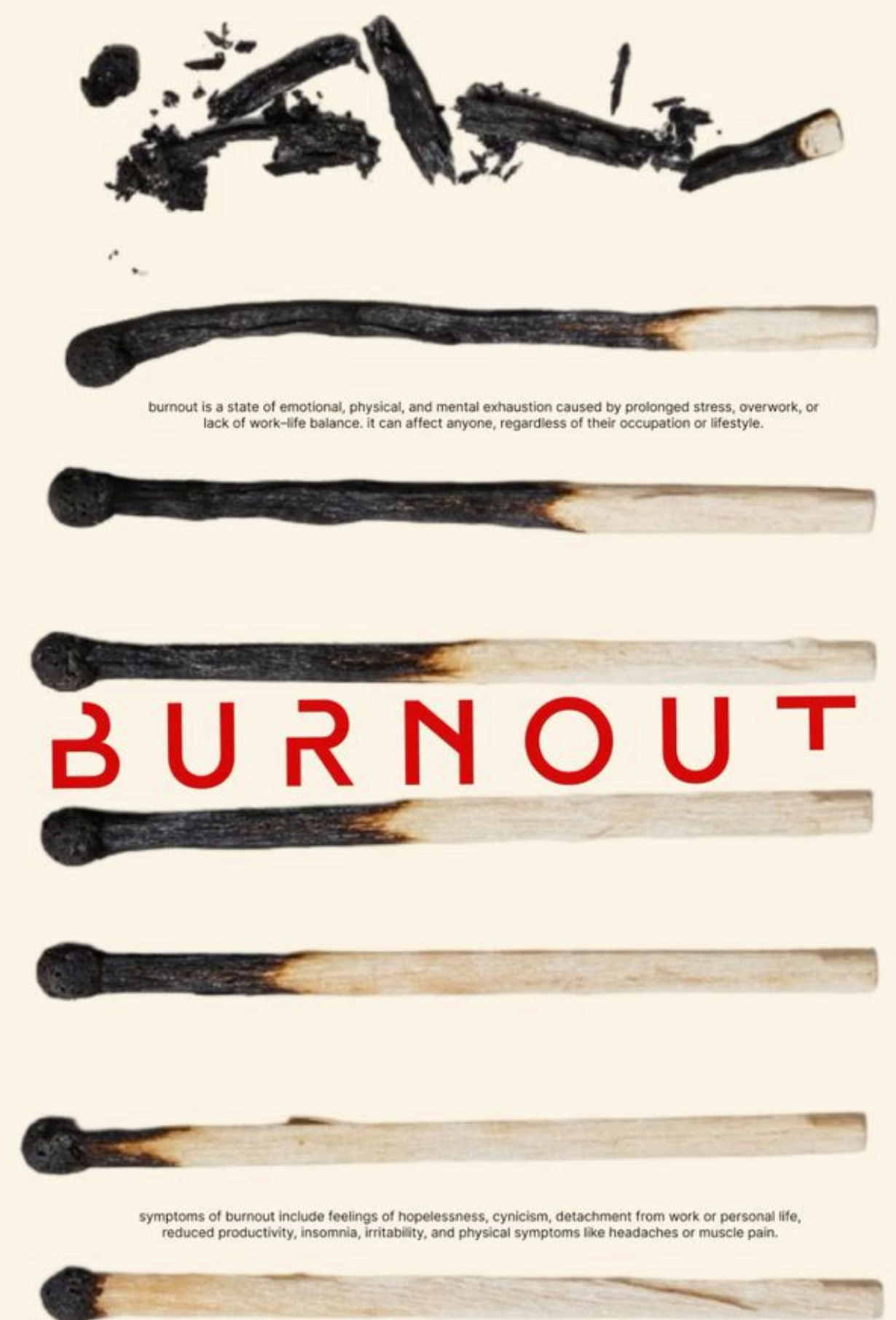


DATA MINING PROJECT

Predicting Employee Burnout in the Tech Industry

A Multi-Model Machine Learning Approach



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Supervised by: Dr. Aloui Donia | Tunis Business School | BA360 Data Mining | 2025–2026

Team Members



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Problem Statement & Context

Why Burnout Matters

#1

Cause of tech
staff turnover

\$\$\$

Cost: absenteeism
& productivity loss

- WHO defines burnout as a syndrome from chronic, unmanaged workplace stress
- Tech industry burnout rates consistently exceed all other sectors
- Consequences: cognitive decline, disengagement, high staff turnover
- Traditional HR reporting misses early warning signals
- Machine learning can detect patterns invisible to human analysts

Project Objectives

PRIMARY OBJECTIVES

Build & Evaluate

Train ML classification models to predict burnout level (Low / Medium / High)

Identify Drivers

Uncover key burnout predictors through feature importance and correlation analysis

Actionable Insights

Deliver business-oriented recommendations for HR strategy and wellness programs

SECONDARY OBJECTIVES

- Compare 5 classifiers: Logistic Regression, Decision Tree, Random Forest, KNN, Naive Bayes
- Apply PCA dimensionality reduction and K-Means clustering for deeper profiling
- Demonstrate sound ML practices: leakage prevention, proper encoding, scaling

Research Questions

RQ1

Which factors most strongly predict burnout?

Feature importance: stress, depression, work hours, anxiety, balance

RQ2

Can ML models accurately classify burnout level?

Low / Medium / High classification from behavioral & psychological data

RQ3

Which algorithm performs best — does complexity pay off?

Benchmark: is ensemble complexity justified over simpler models?

RQ4

Do burnout profiles form distinct clusters in feature space?

PCA visualization: are high/low burnout employees visually separable?

RQ5

How much do support factors moderate burnout risk?

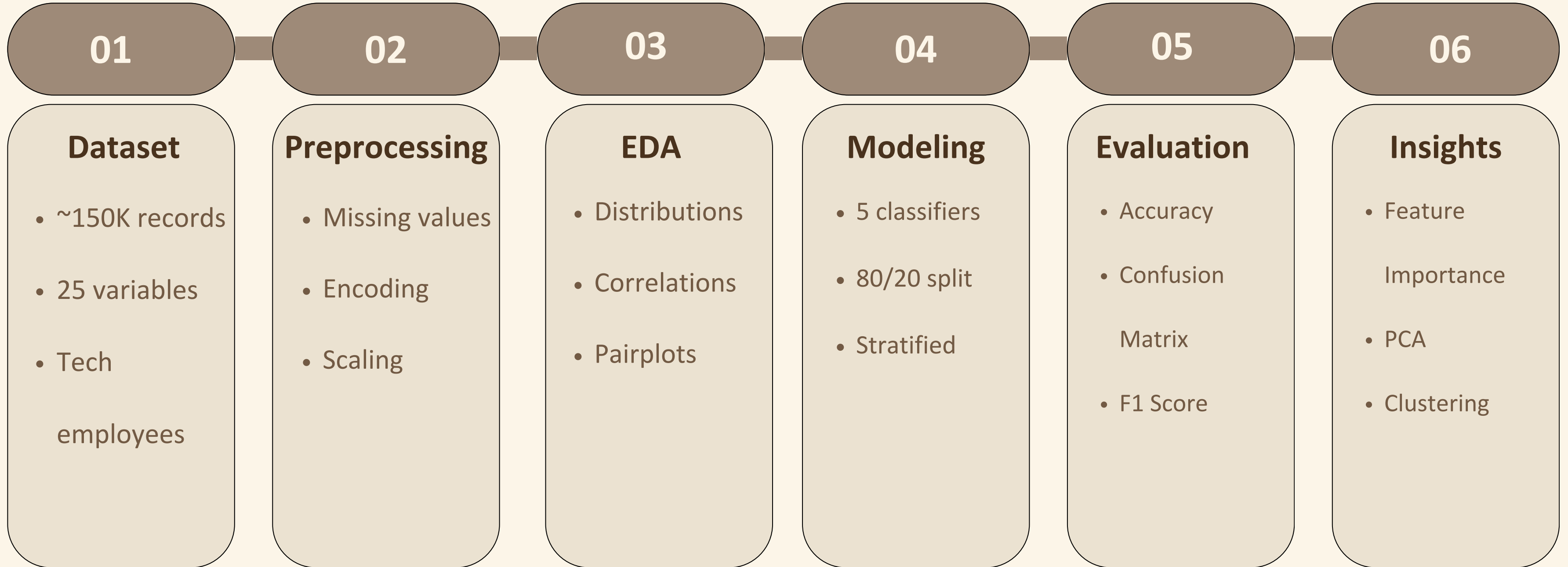
Manager support, social support, work-life balance as protective factors

RQ6

Is burnout a demographic or structural problem?

Age, gender, role vs. working conditions — who is truly at risk?

Methodology



Full ML Pipeline: from raw data to actionable HR recommendations

Dataset Description

~150,000

Employee Records

25

Variables

4

Variable Domains

3

Target Classes
(low / medium / high)

Demographic

- Age (22–55)
- Gender
- Job Role
- Experience Years

Organizational

- Work Hours/Week
- Manager Support
- Work-Life Balance
- Job Satisfaction

Behavioral

- Sleep Hours
- Physical Activity
- Screen Time
- Social Support Score

Psychological

- Stress Level (1–10)
- Anxiety Score
- Depression Score
- ★ Burnout Level
(target)

Data Preprocessing

1 Missing Value Treatment

- Numerical variables (e.g., stress_level, work_hours_per_week) were imputed using the median to avoid distortion from extreme values.
 - Categorical variables (e.g., job_role, work_mode) were filled using the mode.
- This approach allowed us to retain the full dataset (~150,000 observations) without introducing bias.

2 Categorical Encoding

- Variables such as gender, job_role, company_size, work_mode were encoded using Label Encoding.
 - This was suitable for:
 - Tree-based models (Decision Tree, Random Forest)
 - Distance-based models after scaling
- Encoding ensured all features were numerically usable across models.

3 Feature Scaling (StandardScaler)

- Applied to Logistic Regression, KNN, and Naive Bayes.
 - Transforms variables to zero mean and unit variance.
 - Essential because:
 - KNN relies on distance calculations
 - Logistic Regression requires stable optimization
- Not applied to tree-based models, as they are scale-independent.

4 Leakage Prevention (Critical)

- The variable burnout_score was removed from the feature set.
 - It has a strong correlation with the target (burnout_level, $r \approx 0.71$).
 - Keeping it would lead to:
 - Artificially high accuracy
 - A model that cannot generalize in real-world scenarios
- This step ensures the model uses only realistic input variables (behavioral, psychological, organizational).

Machine Learning Models

Logistic Regression (Baseline Model)

- Used to test whether burnout can be predicted through a linear relationship.
- Achieved one of the highest accuracies (~93%), indicating strong linear separability.
- Provides interpretable coefficients, helping identify key drivers such as stress and work hours.

Random Forest (Main Model)

- Ensemble of 100 decision trees trained on bootstrapped samples.
- Reduces variance and improves robustness.
- Provides feature importance analysis, which was crucial to identify:
 - stress_level (most important)
 - depression_score
 - work_hours_per_week
- Achieved performance comparable to Logistic Regression (~93%).

Machine Learning Models

Decision Tree

- Captures non-linear interactions between variables (e.g., stress × work hours).
- Depth limited to 5 to prevent overfitting.
- Offers clear decision rules, making it easy to interpret from an HR perspective.
- Slightly lower performance due to limited complexity.

K-Nearest Neighbors

- Classifies employees based on similarity in feature space.
- Uses $k = 5$ neighbors.
- Requires scaled data due to distance-based nature.
- Confirms that employees with similar profiles tend to share similar burnout levels.
- Computationally expensive for large datasets.

Machine Learning Models

Naive Bayes (Probabilistic Baseline)

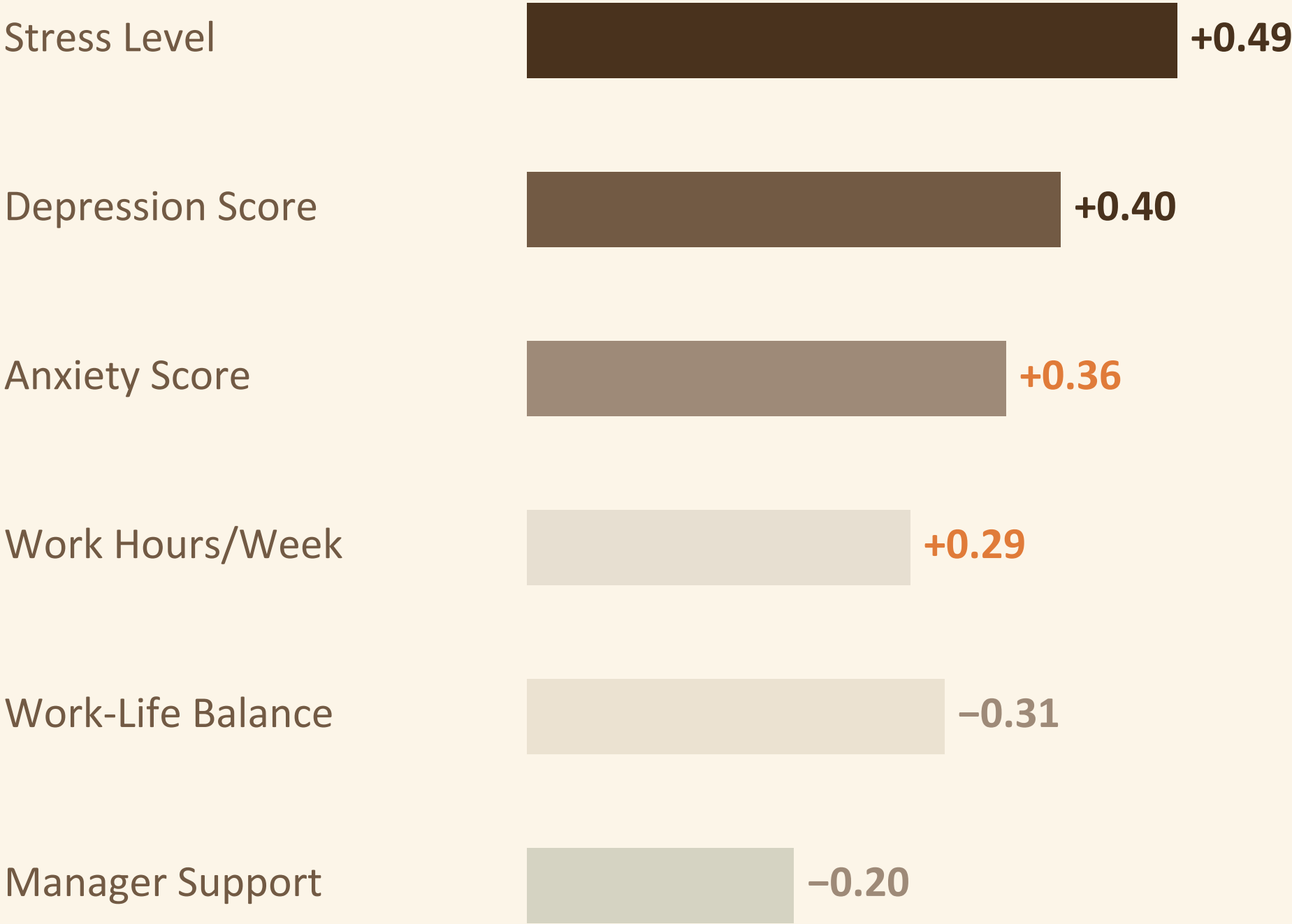
- Based on Bayes theorem with independence assumption.
- Fast and efficient for large datasets.
- Serves as a benchmark model, though slightly lower accuracy (~89%).
- Performs well despite strong simplifying assumptions.

Model Comparision Insight

- All models achieved similar performance (89%–93%), indicating that the dataset contains a strong predictive signal.
- The close performance between Logistic Regression and Random Forest suggests that burnout patterns are largely linear, despite the complexity of the data.

Exploratory Data Analysis — Key Findings

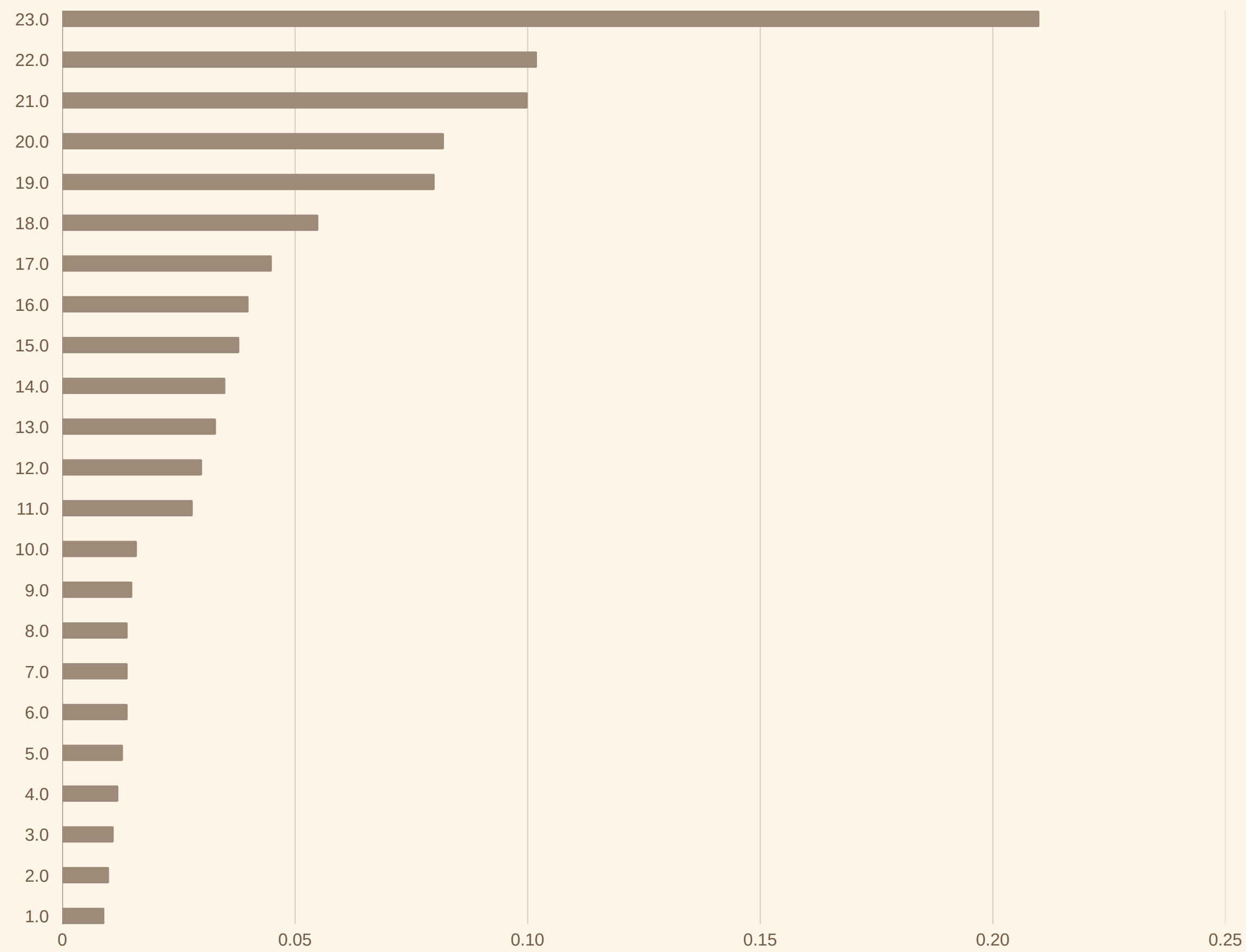
TOP CORRELATIONS WITH BURNOUT LEVEL



KEY EDA INSIGHTS

- Burnout score is right-skewed — most employees score 1–2; high burnout is a meaningful tail
- Stress level alone explains ~21% of Random Forest predictive power
- Employees working 60+ hrs/week are over-represented in high burnout category
- Age, gender, job role: near-zero correlations — burnout is structural, not demographic
- Work-life balance and manager support act as protective buffers

Results — Feature Importance (Random Forest)



TOP 5 PREDICTORS

- 1

Stress Level
Importance: 0.21
- 2

Depression Score
Importance: 0.10
- 3

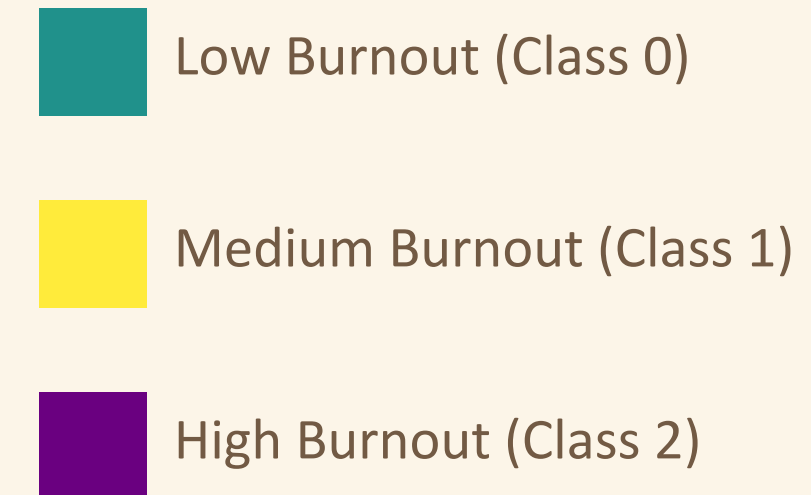
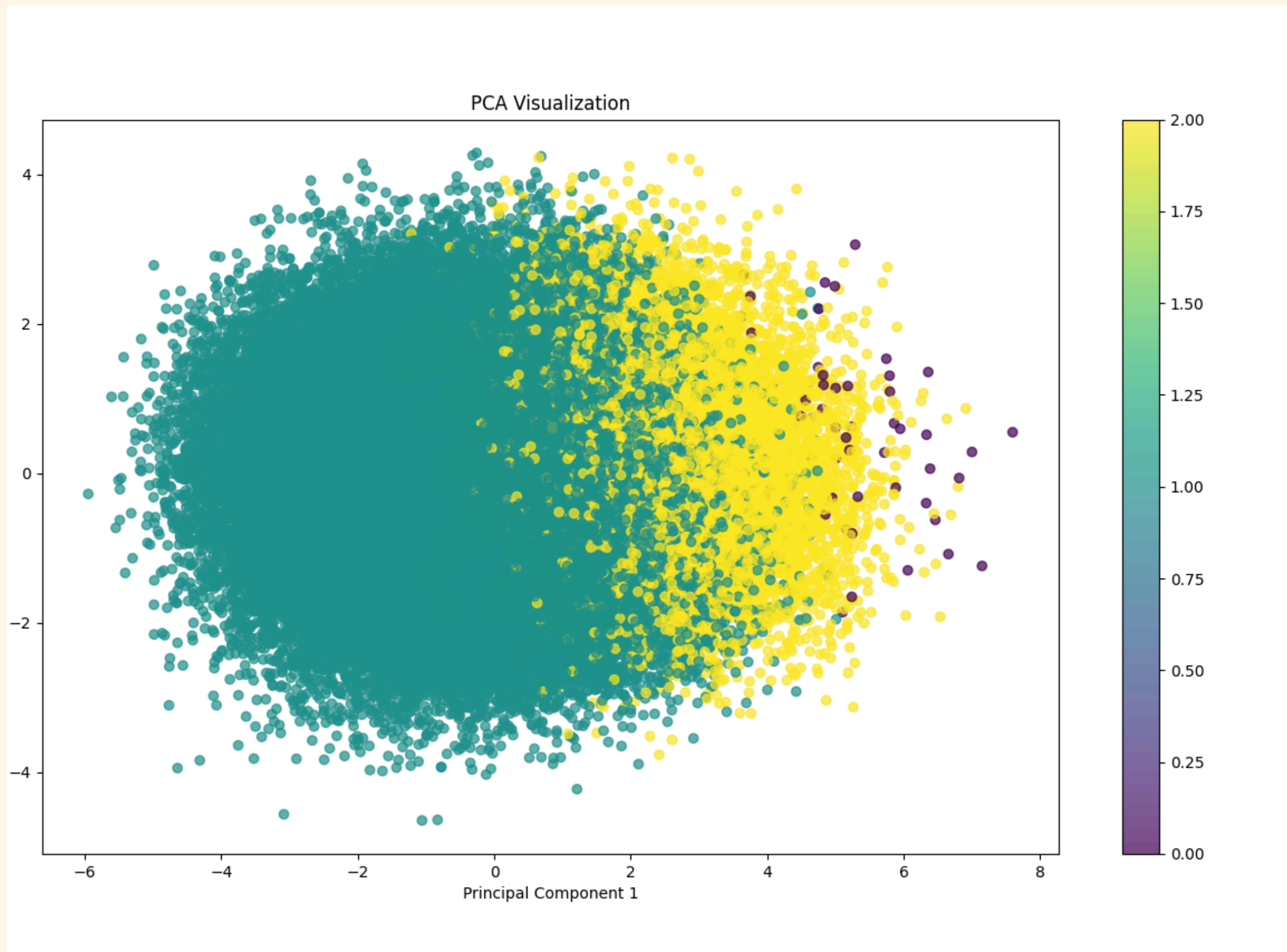
Work Hours/Week
Importance: 0.10
- 4

Anxiety Score
Importance: 0.08
- 5

Work-Life Balance
Importance: 0.08

Demographics rank at the bottom — burnout is a structural, not individual, issue

Results — PCA Visualization

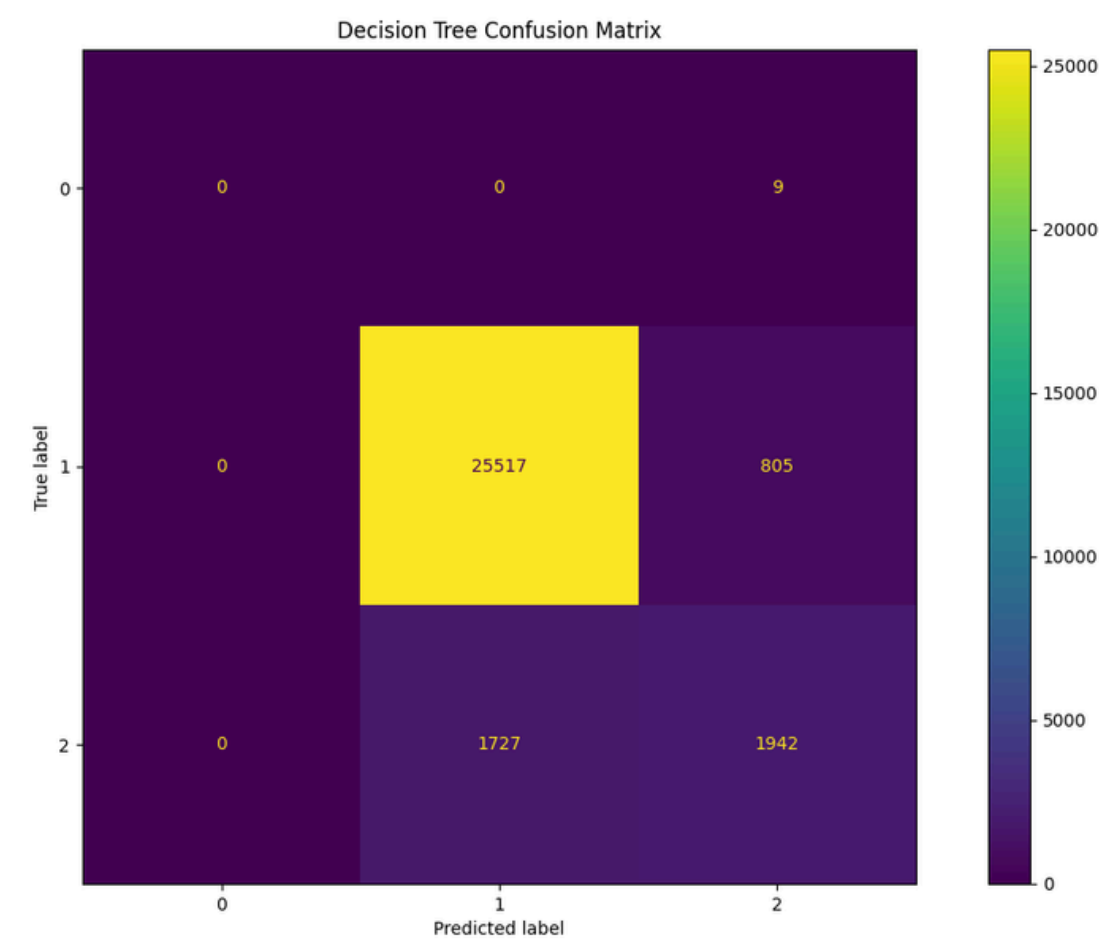


WHAT THIS TELLS US

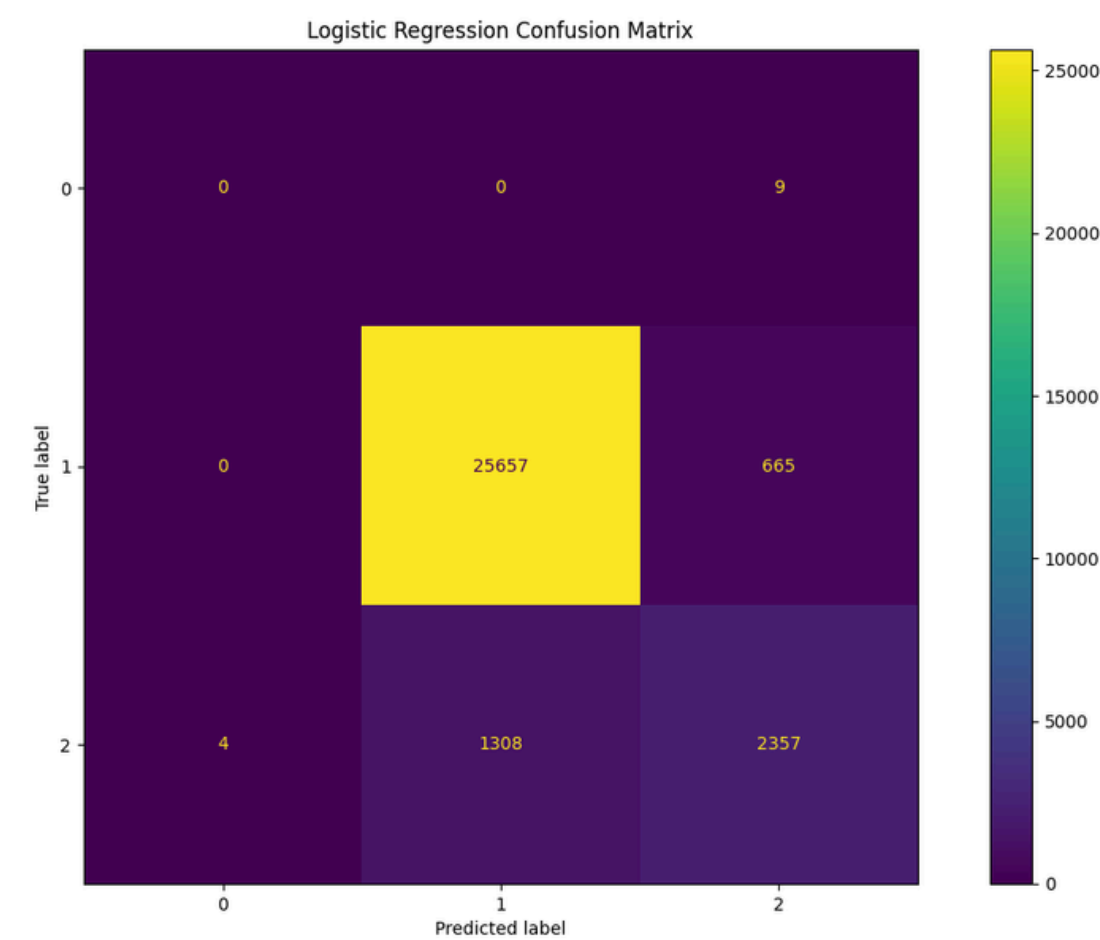
- Low-burnout employees cluster on the LEFT side of PC1
- High-burnout employees cluster on the RIGHT side
- Even in 2D, classes are visually separable
- Confirms our data has genuine discriminative structure
- Explains why Logistic Regression performs on par with Random Forest — the boundary is approximately linear

Confusion Matrices

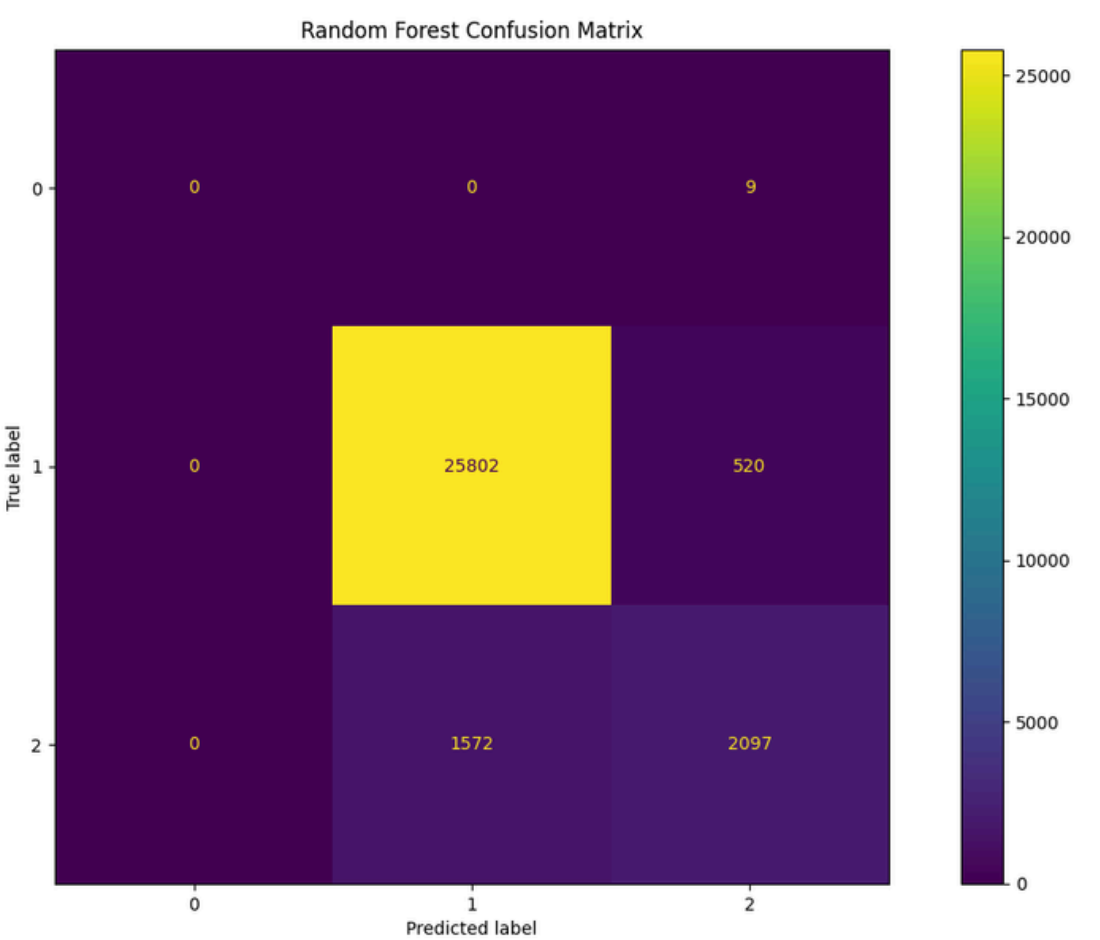
Decision Tree
Confusion Matrix



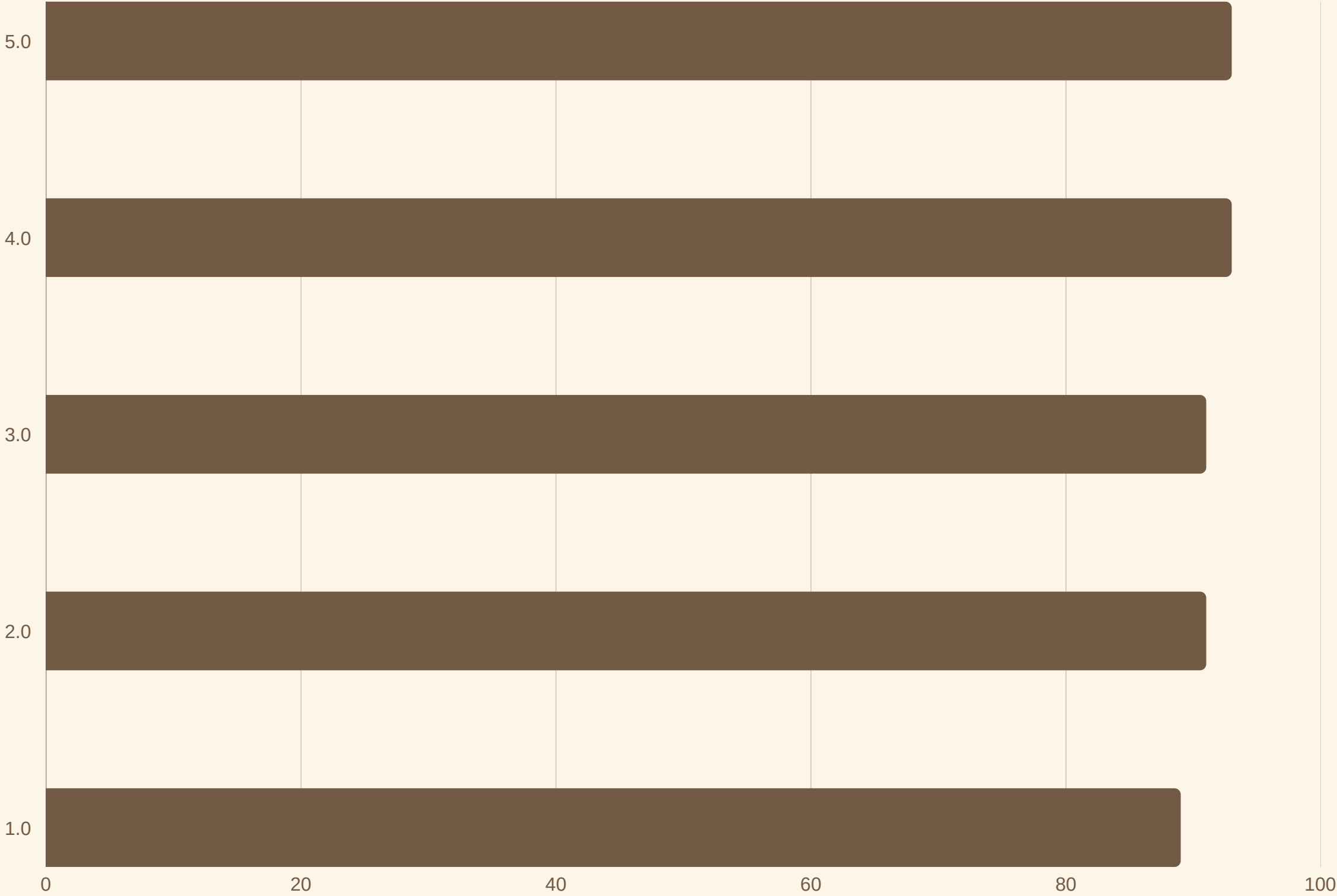
Logistic Regression
Confusion Matrix



Random Forest
Confusion Matrix



Model Comparison & Performance



Model	Accuracy	Strength
Logistic Regression	~93%	Best accuracy + explainability
Random Forest	~93%	Best feature importance
Decision Tree	~91%	Most interpretable rules
KNN	~91%	No training phase needed
Naive Bayes	~89%	Fastest probabilistic baseline

Key Finding: Logistic Regression matches Random Forest — the burnout decision boundary is approximately LINEAR. Simpler models are deployment-ready.

Key Insights

01 Psychology Drives Burnout

0.21

Stress level alone accounts for 21% of model predictive power. Mental health investment = highest ROI.

02 Work Hours & Balance Are Levers

60+ hrs/week

Employees working **60+ hrs/week** are disproportionately high-burnout. Work-life balance has a strong negative correlation ($r = -0.31$).

03 Managers Are Protective

$r = -0.20$

The more support employees get from their manager, the lower their burnout. Among all support-related factors, manager support had the strongest effect ($r = -0.20$)

04 Burnout Is NOT Demographic

$r \approx 0$

Age, gender, and job role show near-zero correlations. Burnout is structural driven by conditions, not who you are.

05 Distinct Burnout Profiles Exist

PCA shows **3** clearly separated **groups** of employees. This means we can design specific programs for each group instead of treating everyone the same

06 Simpler Models Work Fine Here

93% accuracy

Logistic Regression performs just as well as Random Forest. In HR, the simpler and more explainable model is the better choice.

Recommendations for Organizations

Act Now

Real-Time Risk Scoring

Connect the model to your HR system. Get a risk score for each employee before burnout shows up as absence or resignation

Invest in Mental Health

Stress is the **#1 one** predictor. Give employees real access to mental health support, not just awareness campaigns.

Monitor Working Hours

Set an alert when someone consistently works more than 55 hours a week. Give managers the tools to step in before it is too late.

Build over time.

Develop Better Managers

Train managers to spot early signs of stress and have the right conversations. Make this part of their performance review

Cluster-Based Wellness

Use K-Means profiles to design targeted programs per employee archetype . Not one generic wellness initiative for everyone.

Retrain the Model Regularly

Burnout dynamics evolve. Retrain quarterly with updated data. Track recall on the high-burnout class as key KPI.

Limitations & Improvement Pathways

CURRENT LIMITATIONS

- Our data is simulated it is a strong proof of concept, but the real next step is testing it on actual company data.
- We used a simple encoding method to keep things manageable A more precise encoding method would be used in a real deployment setting
- Class imbalance (very few low-burnout cases) model recall on minority class needs improvement
- We worked with data from a single point in time the next version would track employees over time to catch burnout earlier.

IMPROVEMENT PATHWAYS

- Apply tuning for Random Forest and Decision Tree
- Integrate time-series survey data to model burnout trajectory over time
- Apply SMOTE or class weighting to improve high-burnout recall
- Validate on real organizational data for external generalizability

Conclusion

What We Demonstrated

- 5 ML models trained on 23 features : 89–93% test accuracy
- Logistic Regression & Random Forest tied at ~93% accuracy
- Stress level, depression, work hours are primary burnout drivers

The Big Picture

Burnout is expensive for employees and for companies. This project shows that predicting it before it happens is not just possible, it is achievable right now using data that most organizations already have.

Future Work

Next steps: Time-series trajectory modeling, Validation on real organizational data.

THANK YOU

We welcome your questions

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Supervised by Dr. Aloui Donia | Tunis Business School | BA360 — 2025–2026

Dataset: Kaggle — Tech Mental Health & Burnout Dataset | **Course:** Data Mining (BA360)